Wydział
Budowy Maszyn i Informatyki

Uniwersytet
Bielsko-Bialski

Digital Signal Processing

REPORT

Laboratorium no. 1

Topic: Application of Window Functions in Spectral Analysis

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Kierunek: Cyberbezpieczeństwo II stopnia
Semestr: I zaoczny
Grupa: 2a

Link: https://github.com/ryzykotolerowalne/DSP-lab.git

Objective of the exercise:

In digital signal processing (DSP), analyzing the spectrum of a continuous signal requires capturing a finite number of samples N. This process is mathematically equivalent to multiplying the infinite signal by a time-domain window function.

Using a standard rectangular window often introduces sharp discontinuities at the boundaries of the observation interval. When the signal frequency does not align perfectly with the DFT bins, these discontinuities cause a phenomenon known as spectral leakage, where energy from a single frequency leaks into adjacent frequency bands. To mitigate this effect, non-rectangular window functions (such as Hann, Hamming, or Blackman) are applied to smoothly taper the signal amplitudes to zero at the boundaries, improving the dynamic range at the expense of frequency resolution.

Description of exercise:

The purpose of this laboratory exercise was to analyze the impact of different window functions on spectral resolution and the detectability of weak components.

The experiment was performed using Python and the scipy.fft library based on the parameters of Variant 8:

Signal Composition: A discrete-time signal consisting of three closely spaced sinusoidal components was generated:

$$x[k] = A_1 \sin\left(2\pi \frac{f_1}{f_s} k\right) + A_2 \sin\left(2\pi \frac{f_2}{f_s} k\right) + A_3 \sin\left(2\pi \frac{f_3}{f_s} k\right)$$

Parameters: f_1 = 500 Hz), f_2 = 500.2 5Hz, f_3 = 499.75 Hz, with a sampling frequency f_s = 800 Hz and N = 1800 samples.

Amplitudes: The amplitudes were configured with high contrast to simulate a realistic weak-signal detection scenario A_1 = 3.0, A_2 = 0.3, A_3 = 0.03

Note on Aliasing: Because the input frequencies exceed the Nyquist frequency f_s / 2 = 400 Hz, intentional aliasing occurs, causing the digital spectrum components to appear around 300 Hz.

During the exercise, four window functions were applied to the generated signal: Rectangular, Hann, Hamming, and Blackman. For each windowed signal, the Discrete Fourier Transform (DFT) was computed via the Fast Fourier Transform (FFT) algorithm. The resulting spectra were normalized, converted to a logarithmic scale (dB), and plotted to

compare the trade-offs between main lobe width and sidelobe suppression. Additionally, the Discrete-Time Fourier Transform (DTFT) characteristics of the windows themselves were analyzed using zero-padding.

Plots:

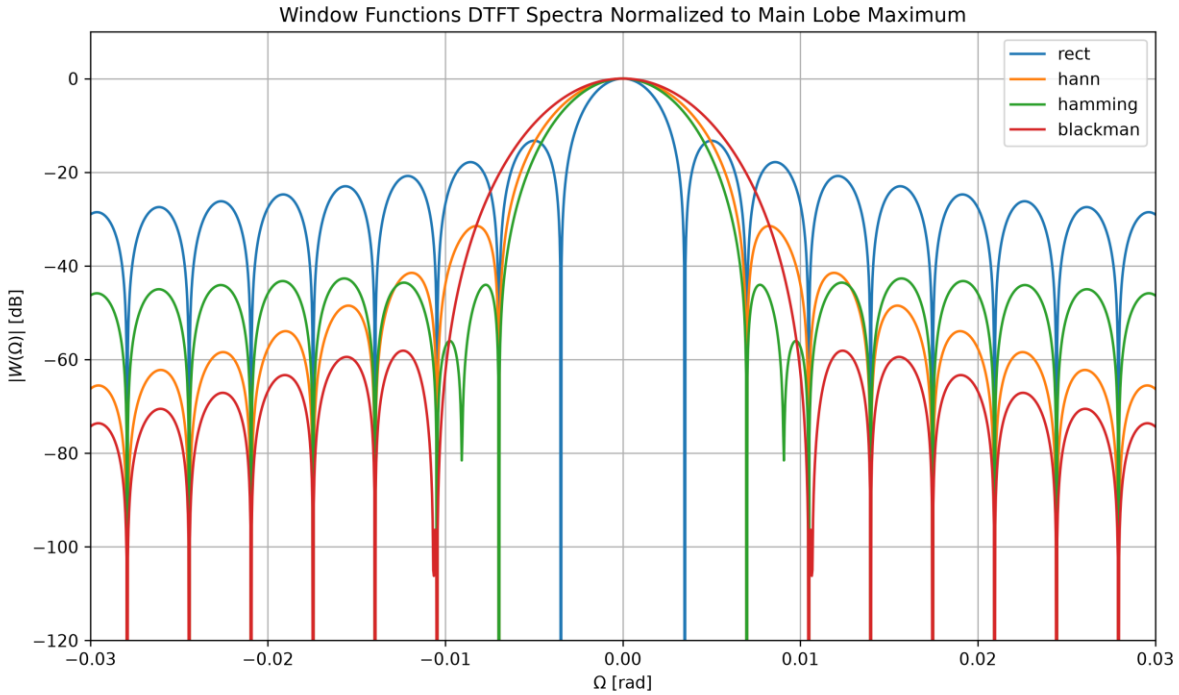


Fig. 1 DTFT Characteristics of window functions.

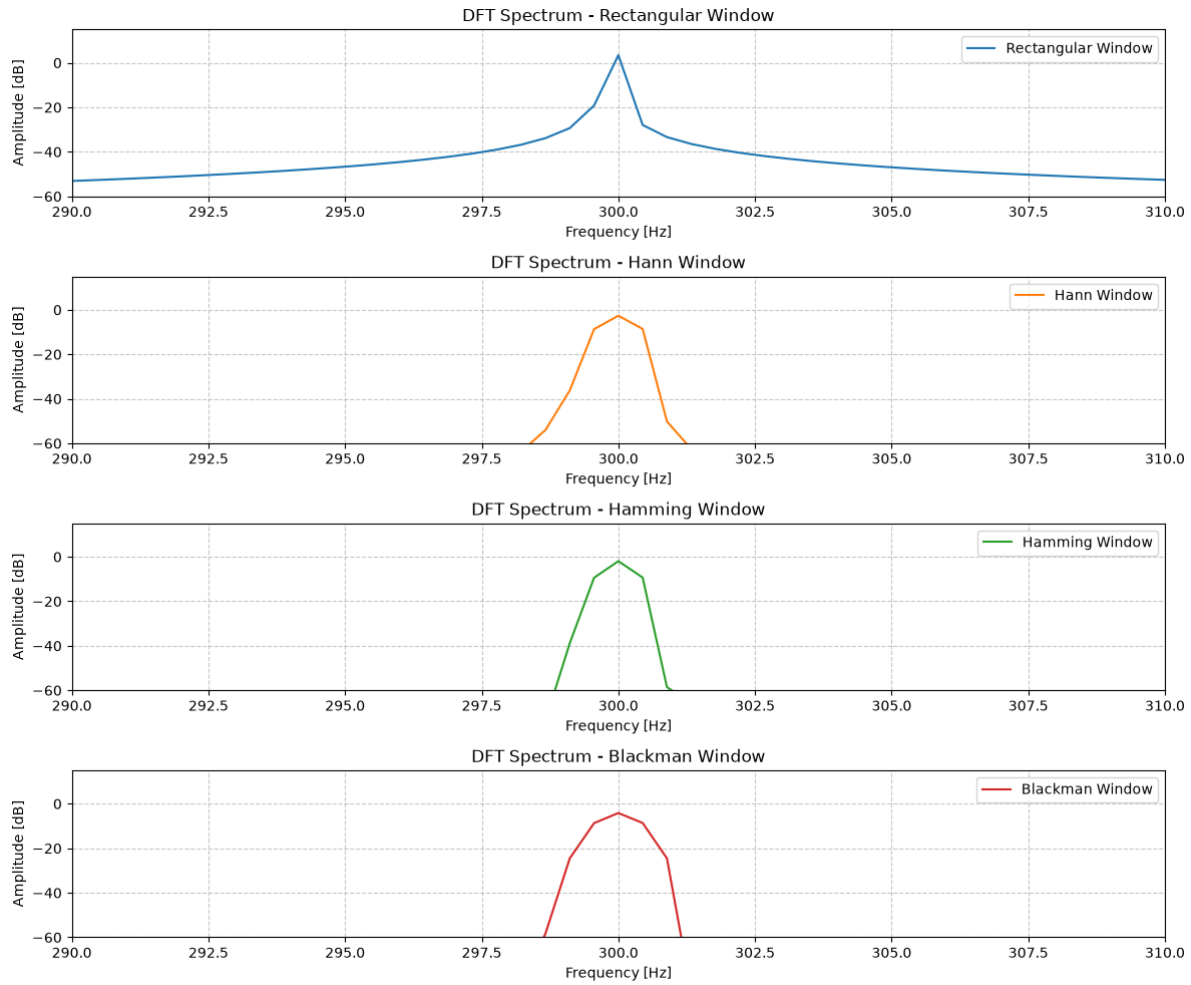


Fig. 2 DFT Spectra of the signal using different windows.

Conclusions:

The Hann and Blackman windows offer the best performance for detecting very weak spectral components (such as f_2 and f_3) in close proximity to a dominant signal (f_1). Due to the severe amplitude differences in Variant 8 ($A_1 = 3.0$, $A_2 = 0.3$, and $A_3 = 0.03$), the weakest tone is attenuated by -40 dB relative to the main tone. The Hann and Blackman windows feature exceptionally low sidelobe levels, which prevents the leakage energy of the strong f_1 component from masking the nearby weak components.

Theoretically, the rectangular window yields the narrowest main lobe, which provides the highest theoretical frequency resolution. However, in this specific measurement scenario, it fails completely due to its high sidelobes causing massive spectral leakage. In terms of a practical compromise between main lobe sharpness and sidelobe suppression, the Hann and Hamming windows provide the best observable separation, allowing the close-by peaks to be distinguished rather than blending into a single smeared peak.

Main Lobe Width: It directly determines the spectral resolution. A narrower main lobe allows the system to distinguish between separate, closely spaced frequency components of

comparable amplitudes. A broader main lobe causes adjacent frequencies to merge into a single broad spectral peak.

Sidelobe Level: It determines the dynamic range and the ability to detect weak components. High sidelobes (as seen in the rectangular window) leak energy into neighboring frequency bins, creating a high floor that drowns out and masks low-amplitude components like f_3. Reducing the sidelobes uncovers these weak tones.

The primary reason for the distinct appearance is the significant difference in their amplitudes ($A_1 > A_2 > A_3$). When performing the DFT on a finite set of samples (N = 1800), the resulting spectrum is a convolution of the signal's ideal frequencies with the spectrum of the chosen window function. Because f_1 is vastly stronger than f_2 and f_3, its main lobe and sidelobe patterns completely dominate the local frequency band, altering or entirely hiding the true shape of the weaker neighboring components depending on the window's properties.